MAT2040 Linear Algebra Final Exam

SSE, CUHK(SZ)

17 Dec 2019

Seat No.: _____ Student ID: _____

i. The exam contains 15 questions.

ii. Put answers in the space after each question. Ask for pages if needed.

iii. Unless otherwise specified, be sure to give **full explanations** for your answers. The **correct reasoning** alone is worth **more credit** than the correct answer by itself.

iv. A table of notations is given in the last page, which you can check-out before the exam.

Question	Points	Score
1	6	
2	6	
3	9	
4	6	
5	6	
6	8	
7	5	
8	6	

Question	Points	Score
9	4	
10	8	
11	4	
12	5	
13	10	
14	7	
15	10	
Total:	100	

Question 1 ..... *6 points*

Judge each of the following statements is TRUE or FALSE in general. Give an example for each statement to support your judgement.

(a) (2 points) A set of orthogonal vectors is linearly independent.

(b) (2 points) For a linear transformation L, if vectors $\mathbf{x}$ and $\mathbf{y}$ are linearly independent, so is $L(\mathbf{x})$ and $L(\mathbf{y})$.

(c) (2 points) The rank of a matrix A is equal to the number of non-zero eigenvalues of A.

Question 2 ..... *6 points*

Let $\mathcal{V}$ and $\mathcal{W}$ be vector spaces with bases $\mathcal{E}$ and $\mathcal{F}$, respectively. Let $L : \mathcal{V} \to \mathcal{W}$ be a linear transformation with the matrix representation A with respect to $\mathcal{E}$ and $\mathcal{F}$, i.e., for any $\mathbf{x} \in \mathcal{V}$,

$$[L(\mathbf{x})]_{\mathcal{F}} = A[\mathbf{x}]_{\mathcal{E}}.$$

Show that

(a) (3 points) A vector $\mathbf{v} \in \mathcal{V}$ is in the kernel of L if and only if $[\mathbf{v}]_{\mathcal{E}} \in \text{Null}(A)$; and

(b) (3 points) A vector $\mathbf{w} \in \mathcal{W}$ is in the range of L if and only if $[\mathbf{w}]_{\mathcal{F}} \in \text{Col}(A)$.

Question 3 ..... *9 points*

Let

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} -3 \\ -1 \end{bmatrix} \tag{1}$$

and

$$\mathbf{v}_1 = \begin{bmatrix} -5 \\ 5 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 4 \\ 3 \end{bmatrix} \tag{2}$$

and let L be a linear operator on $\mathbb{R}^2$ whose matrix representation with respect to the basis $\{\mathbf{u}_1, \mathbf{u}_2\}$ is $A = \begin{bmatrix} 3 & 4 \\ 3 & 2 \end{bmatrix}$.

(a) (2 points) Determine the transition matrix from the basis $\{\mathbf{v}_1, \mathbf{v}_2\}$ to the basis $\{\mathbf{u}_1, \mathbf{u}_2\}$.

(b) (3 points) Find the matrix representation of L with respect to $\{\mathbf{v}_1, \mathbf{v}_2\}$.

(c) (4 points) Let $\mathbf{x}_0 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ and for $k \geq 1$, $\mathbf{x}_k = L(\mathbf{x}_{k-1})$. Calculate $\mathbf{x}_{99}$. (It is okay to keep the power of a scalar in your answer, e.g., 7^{100}.)

Question 4 ..... *6 points*

Consider the following given data points

x	-2	-1	0	1	2
y	4	1	0	1	4

(a) (3 points) Find the best least squares fit by a linear function

$$y = c_0 + c_1 x.$$

(b) (3 points) Find the best least squares fit by a quadratic function

$$y = c_0 + c_1 x + c_2 x^2.$$

Question 5 ..... *6 points*

Let

$$A = \begin{bmatrix} 1 & 3 & 4 \\ 1 & 2 & 1 \\ 3 & 5 & 0 \\ 2 & 4 & 2 \end{bmatrix}.$$

(a) (4 points) Find an orthonormal basis of $\mathrm{Col}(A)^{\perp}$.

(b) (2 points) Calculate the projection of $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 1 \\ -1 \end{bmatrix}$ on to $\mathrm{Col}(A)^{\perp}$.

Question 6 ..... *8 points*

Given the vector space $C[-1,1]$ with inner product

$$\langle f, g\rangle = \int_{-1}^{1} f(x)g(x)dx$$

and norm

$$\|f\| = \sqrt{\langle f, f\rangle}.$$

(a) (1 point) Show that the vectors 1 and x are orthogonal, where 1 is regarded as the function takes value 1 for all inputs in $[-1,1]$.

(b) (3 points) Compute $\|1\|$, $\|x\|$ and $\|x+1\|$.

(c) (1 point) Compute $\cos(\theta)$ where θ is the angle between x and $1+x$.

(d) (3 points) Find the best least squares approximation to $x^{1/3}$ on $[-1,1]$ by a linear function $l(x) = c_0 1 + c_1 x$.

Question 7 ..... *5 points*

Let

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\} \qquad \mathcal{C} = \left\{ \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 5 \\ 2 \end{bmatrix} \right\}$$

be bases of $\mathcal{R}^2$. Find the coordinates of the vector $\begin{bmatrix} 4 \\ -1 \end{bmatrix}_{\mathcal{B}}$ in the standard basis and $\mathcal{C}$.

Question 8 ..... *6 points*

Let A and B be $n \times n$ real matrices. Prove the following statements.

(a) (3 points) If $AB = \mathbf{0}$, then $\text{rank}(A) + \text{rank}(B) \leq n$; and

(b) (3 points) $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$.

Question 9 ..... *4 points*

For any inner product space $\mathcal{V}$, show that the function $\|\mathbf{x}\|$ defined as $\sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}$ satisfies the triangular inequality, i.e., for any $\mathbf{x}, \mathbf{y} \in \mathcal{V}$,

$$\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|.$$

Question 10 ..... *8 points*

Consider the quadratic form $f(x, y, z) = 2xy + 2xz + 2yz$.

(a) (2 points) Find the matrix A associated with the above quadratic form.

(b) (1 point) Show that -1 is an eigenvalue of the matrix A obtained above.

(c) (5 points) Change the variables to obtain a quadratic form associated with a diagonal matrix.

Question 11 ..... *4 points*

Let C be a symmetric positive definite matrix.

(a) (2 points) Show that C is invertible.

(b) (2 points) Show that C^{-1} is also positive definite.

Question 12 ..... *5 points*

The singular-value decomposition (SVD) of an $m \times n$ matrix A with $m \geq n$ is a factorization of the form $U\Sigma V^{\mathrm{T}}$ where

- U is an $m \times m$ orthogonal matrix;
- V is an $n \times n$ orthogonal matrix;
- Σ is an $m \times n$ matrix whose off-diagonal entries are all 0's, and whose (i, i) entries σ_i satisfy $\sigma_1 \geq \sigma_2 \geq \cdots \sigma_n \geq 0$.

Justify the following matrix equalities. Please include detailed intermediate steps of matrix operations.

(a) (2 points) $A\mathbf{v}_j = \sigma_j \mathbf{u}_j$, $j = 1, \ldots, n$.

(b) (3 points) $A = \sum_{i=1}^{n} \sigma_i \mathbf{u}_i \mathbf{v}_i^{\mathrm{T}}$.

Question 13 ..... *10 points*

Consider the matrix

$$A = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}.$$

(a) (7 points) Find the singular-value decomposition of A.

(b) (3 points) Give an orthonormal basis for each of the following subspaces: $\mathrm{Col}(A)$ and $\mathrm{Col}(A^{\mathrm{T}})^{\perp}$.

Question 14 ..... *7 points*

Let A be an $m \times n$ real matrix with $m \leq n$. Denote by $p_1(\lambda)$ the characteristic polynomial of AA^{T} and $p_2(\lambda)$ the characteristic polynomial of $A^{\mathrm{T}}A$.

(a) (3 points) Show the following two matrices are similar

$$M = \begin{bmatrix} \mathbf{0}_{n\times n} & \mathbf{0}_{n\times m} \\ A & AA^{\mathrm{T}} \end{bmatrix}, \qquad N = \begin{bmatrix} A^{\mathrm{T}}A & \mathbf{0}_{n\times m} \\ A & \mathbf{0}_{m\times m} \end{bmatrix}.$$

Hint: construct an invertible matrix B such that $BM = NB$.

(b) (4 points) Show that

$$p_2(\lambda) = \lambda^{n-m} p_1(\lambda).$$

Hints: one approach is to consider the characteristic polynomials of M and N.

Question 15 ..... *10 points*

(**Cayley transform**) For an $n \times n$ matrix S that is real and skew-symmetric, i.e. $S^{\mathrm{T}} = -S$, prove the following statements:

(a) (4 points) Each eigenvalue of S is either 0 or a purely imaginary number. (Hint: a non-zero number c is purely imaginary if $c = -\overline{c}$.)

(b) (3 points) $I + S$ is non-singular.

(c) (3 points) $(I - S)(I + S)^{-1}$ is an orthogonal matrix.

Hints: You can use the following formula without proof, though the proof is rather straightforward:

- $(I + S)(I - S) = (I - S)(I + S)$ and
- $\left(A^{\mathrm{T}}\right)^{-1} = (A^{-1})^{\mathrm{T}}$ where A is an invertible square matrix.

This page has no questions, but a table of notations.

Table of Notations

$\mathbb{R}$	the set of real numbers Without otherwise specified, all matrices have entries from $\mathbb{R}$
$\mathbb{R}^n$	the set of all (column) vectors of n entries from $\mathbb{R}$ the n-dimensional Euclidean vector spaces
$\overline{\lambda}$	complex conjugate. If $\lambda = a + bi$, where a and b are real, $\overline{\lambda} = a - bi$
$\overline{A}$	the complex conjugate of a matrix A, i.e., $\overline{A} = (\overline{a_{ij}})$ for $A = (a_{ij})$
$\mathbf{0}$	the zero vector or the all zero matrix, whose size is implied in the context or specified in the subscript
I_n	the $n \times n$ identity matrix
A^{T}	the transpose of matrix A
A^H	the conjugate transpose of a complex matrix A, i.e., $A^H = \overline{A}^{\mathrm{T}}$
$\det A$, $\det(A)$	the determinant of matrix A
$\mathrm{Col}A$, $\mathrm{Col}(A)$	the column space of matrix A
$\mathrm{Null}A$, $\mathrm{Null}(A)$	the null space of matrix A
$\langle \mathbf{x}, \mathbf{y} \rangle$	the inner product (scalar product) of vectors $\mathbf{x}$ and $\mathbf{y}$
$\|\mathbf{x}\|$	the norm of vector $\mathbf{x}$
$\dim V$, $\dim(V)$	the dimension of a vector space V
$V^\perp$	the orthogonal complement of subspace V
$[\mathbf{x}]_{\mathcal{B}}$	the coordinate vector of $\mathbf{x}$ with respect to the basis $\mathcal{B}$
$C[a, b]$	the set of all the continuous functions defined on the closed interval $[a, b]$